

Scrum Project Implementation

Q1. Write Agile Manifesto.

The Agile Manifesto, foundational to the Scrum Foods project, outlines four core values and twelve principles that prioritize customer value, adaptability, and team collaboration over rigid processes.

Four Core Values:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

Twelve Principles:

- Our highest priority is to satisfy the customer through early and continuous delivery of valuable software
- Welcome changing requirements, even late in development. Agile processes harness change for the customer's competitive advantage
- Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale
- Business people and developers must work together daily throughout the project
- Build projects around motivated individuals. Give them the environment and support they need, and trust them to get the job done
- The most efficient and effective method of conveying information to and within a development team is face-to-face conversation
- Working software is the primary measure of progress
- Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely
- Continuous attention to technical excellence and good design enhances agility
- Simplicity—the art of maximizing the amount of work not done—is essential
- The best architectures, requirements, and designs emerge from self-organizing teams
- At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly

Q2. Write minimum 40 User stories and their Acceptance Criteria along with their BV and CP

User Stories

Q3. What is epic? Write 2 epics.

An epic is a large, complex user story that is too big to complete in a single sprint; it represents a high-level feature broken down into smaller, manageable user stories, themes, or tasks during backlog grooming. In Scrum Foods, epics group related stories from the product vision (e.g., ordering flows) for prioritization in the Product Backlog.

Epic 1: Restaurant Ratings and Reviews

As a user, I want ratings and reviews for restaurants on Scrum Foods, so that I can make informed decisions about where to order from.

Acceptance Criteria:

- Users can view average ratings and reviews for each restaurant on the restaurant details page.
- Users can read detailed reviews and comments left by other customers, relevant after sorting by recency and relevance.
- Users can rate a restaurant and leave a review after placing an order.
- Users can edit or delete their own reviews within a specified timeframe.
- Ratings and review system maintains the integrity and authenticity of user feedback.

Epic 2: Scheduled Orders

As a user, I want to schedule orders in advance on Scrum Foods so that I can plan meals ahead of time and avoid last-minute hassles.

Acceptance Criteria:

- Users can select from available delivery time slots during the day, evening, and night when placing a scheduled order.
- Users can browse and order from all available restaurants during the scheduled order.

- Users receive reminders and notifications regarding their scheduled timeframe, including order confirmation.
- Delivery executives receive clear instructions about scheduled orders and prioritize them accordingly.
- System handles scheduled orders seamlessly, ensuring timely delivery and accurate fulfillment.

Q4. What is the difference between BV and CP

BV (Business Value) measures how important a user story is to the business and stakeholders, estimated using Scrum currency notes (Rs 1000, 500, 100, 50, 20, 10) during Product Vision or grooming. Higher BV stories (e.g., Customer Registration BV 500) get priority in Product Backlog regardless of development effort.

Evaluate Business Value (BV):

- Assess measurable business metrics. For example, a user story that enables upselling or introduces a new revenue stream would have higher BV
- Consider long-term alignment of the FoodExpress story with the strategic objectives
- Assign a relative value or weight to each user story based on its potential business impact or value

CP (Complexity Points/Story Points) measures developer effort and technical complexity to deliver the story, estimated via Planning Poker cards, Modified Fibonacci Sequence (0, 1, 2, 3, 5, 8, 13, 20, 40, 100, ?). Used for sprint capacity planning (e.g., team velocity determines how many CP fit in 2-week sprint).

Evaluate Customer Priority (CP):

- Understand the needs, pain points, and expectations of the end users of the FoodExpress application
- Consider the importance of the user story from the customer's perspective to gauge
- Assess the potential impact on enhancing customer experience, addressing critical user needs
- Assign a relative priority or weight to each user story based on its significance to the customer

Aspect	BV (Business Value)	CP (Complexity Points)
Estimated by	Stakeholders/Client	Scrum Developers
Estimation tool	Currency notes	Planning Poker cards
Purpose	Prioritization (Product Backlog order)	Capacity planning (Sprint planning)
Focus	Business importance/revenue impact	Technical effort/complexity
Example	"Customer Login" BV 500 (high revenue)	"Customer Login" CP 2 (simple CRUD)

Q5. Explain about Sprint

A Sprint is a fixed-duration timebox (2 weeks in Scrum Foods project) where the Scrum Team delivers a potentially shippable Product Increment from selected Product Backlog Items (PBIs/user stories).

Sprint Component Definitions (Scrum Foods Context)

- Time-boxed Duration: Fixed, non-extendable period (2 weeks/10 working days) where sprint commitment must be delivered regardless of completion status.
- Defined Goal: Single, clear objective guiding all sprint work (e.g., "Deliver MVP core flows") - scope negotiable, goal fixed.
- Sprint Planning: 2-hour meeting where team selects Product Backlog stories based on velocity, defines sprint goal, breaks stories into tasks, team self-allocates.
- Daily Stand-up Meetings: 15-minute daily synchronization where developers answer 3 questions: What did you do yesterday? What will you do today? Any impediments?
- Development and Testing: Continuous coding, unit testing, integration within sprint; all "Done" items meet Definition of Done (DoD) - coded, tested, demo-ready.
- Incremental Delivery: Potentially shippable Product Increment from all "Done" Sprint Backlog items at sprint end, ready for client demo/deployment.
- Sprint Review: 4-hour meeting with stakeholders to demo increment, discuss velocity/BV delivered, gather feedback, adapt Product Backlog priorities.
- Sprint Retrospective: 1.5-hour team-only meeting to discuss: What went well? What didn't? What to improve? Action items for next sprint.

- Backlog Refinement: Ongoing 10% capacity activity (2 hrs/week) to prepare top Product Backlog items - add acceptance criteria, estimate BV/CP, split epics, ensure DoR met.

What is Sprint

- Duration: 2 Weeks - Your sprint (10 working days)
- Value: Delivers working, tested features (MVP increments)

What is Scrum

- Duration: 1 day
- Scrum is a subunit of Sprint
- Your Scrum Value: Daily progress check and synchronization

Key Rules:

- Fixed timebox (no extension)
- Scope can shrink, never grow
- Team self-organizes task allocation
- Daily transparency via Scrum meetings
- Continuous integration/testing

Q6. Explain Product backlog and sprint back log

Product Backlog

Definition: The Product Backlog is the single, living source of truth containing ALL work to be done on Scrum Foods throughout the entire project lifecycle.

Key Characteristics:

- Owner: Product Owner (YOU as PO)
- Content: 52+ User Stories, Epics, Bugs, Technical Debt, Improvements
- Prioritization: Ordered by Business Value (BV) - Rs 1000 > 500 > 200 > 100 > 50
- Dynamic: Continuously refined through grooming sessions
- Granularity: Mix of Epics (high-level) and Sprint-ready stories (detailed AC, BV, CP)

Sprint Backlog

Definition: The subset of Product Backlog selected for ONE specific sprint (2 weeks), broken down into actionable tasks.

Key Characteristics:

- Owner: Development Team (8 Developers)
- Content: 10 PBIs → 20 Tasks → WIP → Done
- Timeboxed: Fixed 2-week duration
- Frozen Scope: No new stories mid-sprint
- Forecast-Based: Selected based on team velocity

Product Backlog	Sprint Backlog
Anything that the project vision accomplishes	Anything that the sprint goal fulfills
Product Owner owns	A subset of Product Backlog. Development team owns
Contains requirements, defects, tasks	As prioritized by the Product Owner
Everyone contributes to the product backlog refinement meeting	Only the development team contributes
To refine the product backlog items	Sprint planning meeting to refine sprint backlog items
Will be backed by PO through the product lifecycle	Backlog items once sprint has started
Release burndown metric is used	Sprint burndown metric is used
Definition is done at a user story level	Definition is done at the activity or task level
Daily standup backlog items not discussed	Daily standup discusses sprint backlog items once sprint goal

Q7. What is impediments log? write 2 impediments

Impediments Log: All challenges faced by the Scrum Team during sprints are systematically logged in the Impediments Log. The Scrum Master is responsible for tracking, prioritizing, and resolving these blockers to protect team velocity and ensure sprint commitment delivery.

Purpose:

- Daily Scrums: Team raises "Any Challenges/impediments?" → Added to log
- Visibility: Posted publicly (physical board/digital dashboard)
- Prioritization: High-impact blockers first

- Resolution: Scrum Master removes within sprint or escalates to stakeholders
- Retrospective: Analyze patterns for process improvement

Log Id	1
Description	Delivery Partner Shortage in specific regions
Impact	Delays in order delivery and increased customer dissatisfaction
Priority	High due to its impact on customer experience
Status	Open
Assigned to	Operations Team
Action taken	Operations team and HR team onboarding new delivery partners in the region
Resolution	The operations team is working on fast-track onboarding new delivery partners in the region. The HR team streamlining delivery partner recruitment efforts are ongoing

Log Id	2
Description	Payment Gateway API Integration Delay
Impact	Unresponsive payment gateway- Core MVP revenue feature stalled
Priority	Critical - Blocks sprint goal achievement
Status	Resolved
Assigned to	Infrastructure Team
Action taken	Scrum Master escalated to Razorpay vendor support team
Resolution	Sandbox credentials received Day 5. All payment stories unblocked

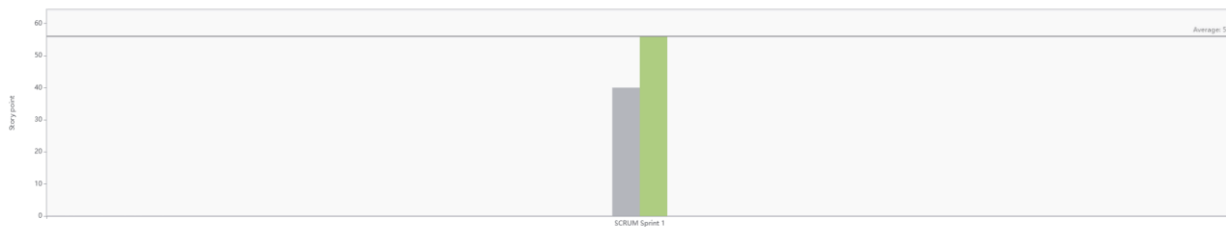
Q8. Explain Velocity of the Team.

Velocity refers to the measure of the amount of work the development team can complete during a sprint. It is performed by the development team itself as they are

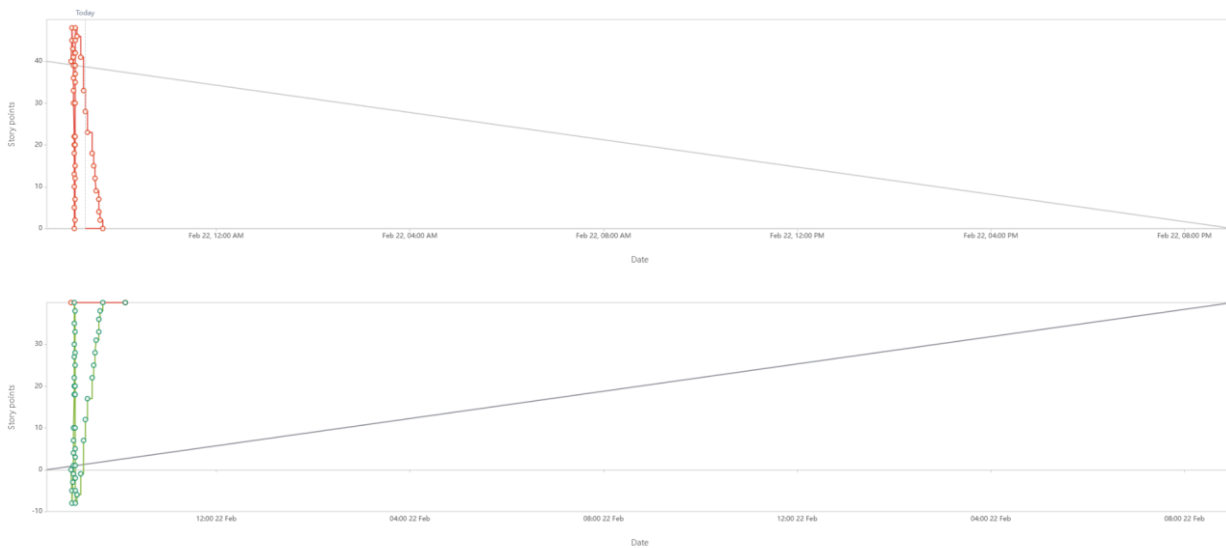
responsible for estimating the effort required to complete each user story or backlog item.

How Velocity is Calculated:

- Story Point Estimation: Team estimates CP (1,2,3,5,8...) during Sprint Planning
- Tracking Completed Story Points Work: Track actual CP done via Sprint Burndown Chart
- Average Velocity Points: 3-sprint rolling average (Sprint 1: 25 CP, Sprint 2: 28 CP → Avg 26.5 CP)



Q9. Draw Sprint Burn Charts and Product Burn Down Charts



The sprint burndown chart was prepared in JIRA with sprint timings (start time and end time) of just half an hour; hence, the chart created would not match a real-life scenario.

Product burndown charts can't be prepared in JIRA; however, the product burndown chart looks similar, wherein the X-axis shows sprints instead of days, and the Y-axis shows remaining story points in the product backlog. Here is the difference between the two:

Aspect	Sprint Burndown Chart	Product Burndown Chart
Scope	Single sprint (10 days, 25 CP)	Entire release (8 sprints, 350 CP)
X-Axis	Days (Day 1-10)	Sprints (S1-S8)
Y-Axis	Remaining story points in sprint backlog	Remaining story points in product backlog
Purpose	Daily team progress tracking	Release completion forecasting
Updated	Daily during sprint	End of each sprint
Audience	Development team, Scrum Master	Product Owner, Stakeholders

Q10. Explain about Product Grooming.

Product grooming, also known as backlog grooming or refinement, is an important activity in Agile software development that involves reviewing, prioritizing, and refining items in the product backlog.

Key Characteristics:

- Goal: Ensure that the backlog is well-prepared, organized, and ready for implementation in upcoming sprints
- Collaborative: Involves Product Owner (YOU), development team, and other relevant stakeholders
- Ongoing: Typically 10% of sprint capacity (2 hours/week in Scrum Foods)
- Focus: Top 15 Product Backlog items must meet Definition of Ready (DoR)

How It Is Conducted

- Setting the Context: Establish meeting purpose and review current sprint status/priorities
- Backlog Review: Examine top 15 Product Backlog items for clarity and relevance
- Prioritization: Reorder stories by BV/CP ratio, MoSCoW, stakeholder input

- Refinement and Estimation: Split epics, write acceptance criteria, Planning Poker for CP
- Dependency Analysis: Identify story interdependencies and sequencing requirements
- Acceptance Criteria: Define testable, clear acceptance criteria for Definition of Ready (DoR)
- Backlog Grooming Meetings: Weekly 1-hour collaborative sessions with PO, team, stakeholders

Q11. Explain the roles of Scrum Master and Product Owner.

Criteria	Product Owner	Scrum Master
Nature of work	Collaborative and brings the vision of all stakeholders into the product backlog	Acts as a coach and is responsible for maintaining the quality of the product
Responsibilities	Responsible for backlog. Intermediary between customers team and the development team	Ensures the Scrum framework is followed and development team creates quality product
Accountability	Backlog, timely completion of stories for stakeholders, and for providing client reports	Accountable for the quality of the entire project and completion of the product
Reporting	Reports to top management and the clients	Reports to top management about the efficiency of the team
Skills	Communication, creativity, critical thinking, and a sharp mind are key assets for Product Owner	Knowledge and practices of Scrum theory. Being able to lead a team without a sense of authority

Q12. Explain all Meetings Conducted in Scrum Project.

- **Sprint Planning:** This meeting occurs at the beginning of each sprint and involves the product owner shares the backlog items. The product owner shares the prioritized backlog items, and the development team collaborates to select the sprint goal and create the sprint backlog, which outlines the specific tasks.

- **Daily Stand-up (Daily Scrum):** This short daily meeting, typically lasting 15 minutes, involves the development team discussing progress since the last stand-up, each team member shares their focus for the day, and challenges or impediments are identified.
- **Sprint Review:** At the end of each sprint, the sprint review meeting takes place. The development team demonstrates the work completed to stakeholders, including the product owner and possibly customers or users. The purpose is to assess progress, gather feedback, and determine potential adjustments to the product backlog.
- **Sprint Retrospective:** Immediately following the sprint review, the sprint retrospective occurs. It focuses on what went well, areas for Scrum team improvement, and continuous learning, adjustment, and enhancing team effectiveness.

Optional meetings:

- **Backlog Grooming (Refinement):** This meeting involves the product owner and development team focusing on upcoming sprints. It ensures that the backlog items are well-prepared and refined.
- **Release Planning:** This meeting occurs at the start of a project or major releases. Stakeholders discuss and plan the high-level scope, timeline.
- **Ad hoc meetings:** These meetings may be scheduled as needed to address specific topics or issues, such as resolving impediments, discussing challenges, or conducting additional planning or collaboration.

Q13. Explain Sprint Size and Scrum Size.

Sprint Size: The sprint size refers to the length or duration of a sprint. A sprint is a time-boxed period during which the development team works to deliver a potentially shippable product increment. Sprint sizes are determined by the project planning phase and typically range from one to four weeks. The most common sprint duration is two weeks. Fixed time-boxed duration of one sprint (Scrum Foods: 2 weeks = 10 working days)

Scrum Team Size: The scrum team size refers to the number of individuals who collaborate to deliver the product increment. Scrum teams are self-organizing and cross-functional, typically consisting of a product owner, scrum master, and development team. The recommended scrum master size is small, ideally between five and nine members. However, there are no strict rules regarding team size, and it may

vary depending on the specific project requirements and complexity. Optimal: 5-9 members (ideal communication, self-organization)

Q14. Explain DOR and DOD

Definition of Ready (DoR)

Definition of Ready (DoR) criteria that a user story must meet before it is considered ready to be included in the sprint. It ensures that the team has sufficient information and clarity about the user story, reducing the likelihood of misunderstandings or delays during the sprint.

DoR Checklist:

- The user story has a clear and concise description, including the expected behavior/functionality
- The acceptance criteria are defined and agreed upon by the product owner and the development team
- Any necessary wireframes related to the user story are appropriately sized or estimated in story points or other relevant units
- Any dependencies or external resources needed for the user story are identified and accessible
- The team has sufficient information and clarity about the user story to begin development

Definition of Done (DoD)

Definition of Done (DoD) criteria that a user story or any backlog item must meet to be considered complete and ready for release. It establishes shared understanding that all necessary aspects, such as quality, testing, and documentation, are addressed.

DoD Checklist:

- The code for the user story is implemented, reviewed, and merged into the main codebase
- Automated tests are created correctly and do not introduce regressions
- The user story is thoroughly tested and validated against the defined acceptance criteria
- The user interface (UI) or user experience (UX) aspects related to the user story are implemented and reviewed
- Release notes are documented, including any relevant instructions, guides
- Product Owner has reviewed and accepted the user story as meeting expected requirements

Q15. Explain Prioritization Techniques and MVP

MoSCoW Technique (Prioritization)

MoSCoW is a prioritization technique that categorizes Product Backlog items into four groups: Must have (critical requirements), Should have (important but not vital), Could have (desirable if time permits), and Won't have (out of scope for current release).

- M - Must Have: Critical business requirements (Delivery Boy Registration and Customer Registration)
- S - Should Have: Important but not critical (Place Order- essential revenue but dependent on login)
- C - Could Have: Nice-to-have if time permits (Order History - enhances UX)
- W - Won't Have: Out of scope for this release (Advanced analytics dashboard - future sprints)

Minimum Viable Product (MVP)

MVP is a product version with the basic and essential features only that delivers customer value, solves the core problem, and enables early feedback collection while minimizing development effort.

- Essential Features (Core/Must-Have):
 - Features absolutely required for MVP to solve the core problem and deliver promised value. Without these, the product fails its purpose.
 - Scrum Foods: Delivery Boy Registration, Customer Login and Place Order - no food delivery without these.
- Basic Features (Minimum/Functional):
 - Features that make essential features usable but aren't the core value themselves. Provides minimal polish/functionality.
 - Scrum Foods: Simple UI forms, basic error messages, database CRUD - makes registration/login work but doesn't generate revenue.

Q16. Difference between Business Analyst n Product Owner

Aspect	Product Owner (PO/BA)	Business Analyst (BA)
Strategic Role	Market Strategy Owner - Defines overall product vision and go-to-market strategy	Requirements Analyst Strategy - Gathers and documents detailed functional requirements

Vision & Planning	Enterprise Vision - Creates high-level product roadmap and release planning	Non-Functional Analysis - Documents technical and performance requirements
Backlog Ownership	Competitive Backlog Analysis - Prioritizes Product Backlog using BV/CP/MoSCoW	User Stories - Writes detailed, INVEST-compliant user stories with acceptance criteria
Stakeholder Management	Stakeholder Strategy - Represents business stakeholders, resolves conflicts	Modeling Focus - Creates UML diagrams, wireframes, process flows
Execution Focus	Go to Market - Ensures business value delivery through sprints	Project Focus - Ensures smooth project execution through requirement traceability
Ceremonies	Stakeholder Management Planning - Participates in Sprint Review, accepts increments	Conflict Resolution - Facilitates grooming, clarifies requirements during Sprint Planning
Key Skills	Backlog Cost/Benefit - Business acumen, prioritization, stakeholder communication	Prioritization of Stories - Analytical skills, documentation, technical understanding

Q17. Prepare a sample Resume of 3yrs exp Product Owner

Rahul Sharma

Product Owner | CSPO Certified | 3+ Years Agile Experience

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Professional Summary

Results-driven Product Owner with 3+ years delivering high-impact MVPs and optimizing product backlogs using MoSCoW prioritization. Delivered 35% revenue growth through 18 successful sprints, 94% on-time delivery, and stakeholder satisfaction scores of 4.8/5. Expert in backlog grooming, sprint planning, and business value maximization.

Professional Experience

Senior Product Owner, TechFlow Solutions, Bengaluru | Jan 2025 – Present

- Led Scrum Foods MVP launch: Prioritized 5 essential stories (*Delivery Bay Raditation BV500, Place Order BV500*) using MoSCoW - *achieved 25 CP sprint velocity, validated core business model*
- Implemented DoR/DoD framework; across 6 sprints, reducing technical debt by 40% and improving release readiness
- Facilitated backlog refinement with 12 stakeholders, resolving 25+ conflicts through business value cost analysis

Product Owner, InnovateX Labs, Bengaluru | Aug 2023 – Dec 2024

- Delivered Order Management System for 50+ restaurant clients: MoSCoW prioritization achieved 92% story completion rate across 12 sprints
- Created INVEST compliant user stories with acceptance criteria, reducing rework

Associate Product Manager, Celonis India, Bengaluru | Jun 2022 – Jul 2023

Key Projects & Achievements

-  **Scrum Foods MVP:** Delivered 5 essential features in 2 weeks
 - Business Value: ₹5L revenue potential validated
 - Team Capacity: 25 CP achieved (94% velocity)
-  **OrderSync Platform:** 50 restaurants onboarded, 35% order efficiency gain
-  **Process Mining Dashboard:** Reduced development time by 40%, 85% user adoption rate

Core Competencies

- Product Strategy | Backlog Management | Agile Execution | MoSCoW Prioritization | DoR/DoD Implementation
- Sprint Planning | MVP Definition | INVEST User Stories | Stakeholder Facilitation
- Business Value Maximization | Release Planning | Conflict Resolution

Technical Skills

- Jira, Confluence, Azure DevOps, Trello | MS Visio, Balsamiq, Lucidchart
- Scrum, Kanban, SAFe | Google Analytics, Mixpanel

Certifications & Education

- Certified Scrum Product Owner (CSPO) - Scrum Alliance, 2024
- Product Management Certification - Product School, 2023
- B.Tech Computer Science - VTU, Bengaluru, 2022 (8.2 CGPA)

Strategic. Analytical. Delivery-Focused.